

Ambivalence and attitudes in consumer replacement decisions

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Abstract

This research examines consumer replacement decisions and the ambivalence that may accompany such decisions. Durable goods replacement decisions are complex in that they involve two interconnected but distinct subdecisions: whether to replace the good, and what to do with the incumbent possession. This research proposes and tests a decision model that incorporates both of these subdecisions. It also examines sources of conflict associated with the ambivalence that may accompany replacement decisions. In so doing, it describes some potential problems associated with conflict measures that have been previously used in the ambivalence literature and describes methods to avoid these problems. Findings from a longitudinal study demonstrate that a model incorporating elements of both replacement subdecisions offers superior prediction of intentions over a traditional decision model. Findings also reveal that ambivalence may have effects at both the pre- and post-purchase stages of the decision process, and that the relationship between conflict and ambivalence may be more complex than previously recognized.

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When consumers contemplate replacing an important possession, they frequently must resolve several issues related to the replacement task. Some of these issues center on the possession the consumer already owns. The decision to replace a car, for instance, is likely to evoke consideration of costs and benefits of keeping or disposing of the vehicle one already owns, evaluation of the adequacy of the decision maker's financial resources and garage space, consideration of the household's short- and long-term consumption goals and values, and a variety of situational influences. Other issues, like an assessment of new models available in the marketplace, are related to decisions about the potential replacement product. To further complicate matters, the consumer may also need to justify the ultimate decision to others who might be affected by it and perhaps involve them in the decision process. In short, replacement decisions involving durable possessions can be quite complex.

This paper examines consumer decisions involving replacement of durable goods. We propose that replacement decisions

are the product of two parallel but integrated suborder decisions involving, on one hand, the continued retention of a currently owned possession and, on the other hand, the acquisition of a replacement product. In the literature, acquisition and disposition have typically been treated as separate processes rather than interconnected events in the lives of consumers (e.g., [Jacoby, Berning, and Dietvorst, 1977](#)). In actuality, however, the decision to replace a consumer durable is often intimately connected with the decision of what to do with an incumbent possession. Therefore, we predict that knowledge of attitudes toward both actions should increase the accuracy of predictions of replacement intentions and actual replacement behavior.

The added complexity of making two decisions as opposed to just one can create opportunities for feelings of ambivalence to arise in replacement decisions. Ambivalent feelings represent a state of psychological conflict concerning an attitude object ([Priester and Petty, 1996](#)), and in replacement decisions, there are many opportunities for conflict to arise. For example, a person could experience conflicting attitudes or feelings about the decision ([Otnes, Lowrey, and Shrum, 1997](#)). If the old car is still in working order, it may feel wasteful to replace it despite the owner's desire for something new. If the car has nostalgic or other personal meanings, or simply has been a pleasant and reliable partner, the owner may be reluctant to dispose of it ([Cripps and Meyer, 1994](#); [Strahilevitz and Loewenstein, 1998](#)), even though a different one may better meet her needs. In

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addition, interpersonal conflict may arise if family members or friends harbor opinions about the replacement decision that conflict with the decision-maker's attitude (Priester and Petty, 2001). In this paper, we explore how these sources of conflict may influence feelings of ambivalence and examine the effects that such ambivalence may have on purchase replacement decision processes.

The objective of this manuscript is to examine the unique characteristics of replacement decisions and the implications of these characteristics for consumer decision processes. We do so by (1) conceptually and empirically examining the interplay of attitudes, ability, and social influences concerning the two, sometimes conflicting, courses of action involved in replacement and by (2) examining the role of ambivalence in the replacement decision process. We propose that inclusion of variables related to both the replacement and retention decisions will enhance prediction of behavioral intentions, and we show that incorporating feelings of ambivalence can increase our understanding of how the replacement decision process proceeds. The manuscript describes a framework for replacement decisions that is based on this expanded perspective and reports a longitudinal study that tests the hypotheses embedded in that framework. A third contribution is to examine potential sources of ambivalence in the replacement decision process. In doing so, we examine the appropriateness of commonly-used measures of ambivalence sources, note the statistical limitations of these measures, and propose a way to circumvent these limitations.

What does it mean to “replace”?

In many instances, the purchase of a durable product is a replacement activity, not a first time purchase experience in the product category. This distinction between replacements versus first-time purchases is one that has been largely overlooked by scholars. We argue, however, that replacement has some unique properties, including the need to make two separate but interrelated decisions and to reconcile sources of conflict that may have a bearing on overall decision outcomes. The few studies that have examined replacement treat it as an acquisition-oriented activity that involves the disposition of an incumbent possession (Bayus, 1991; Okada, 2001). Okada (2001) demonstrated that factors related to incumbent products (i.e., past usage experience, duration of possession) can exert a significant influence on the decision to purchase a replacement product. Nevertheless, in her study and others, disposition seems to be a foregone conclusion. We take a slightly different approach that more accurately represents the reality of consumer replacement decisions.

For purposes of this study, “product replacement” is said to occur when the primary role of a currently owned product in a consumer's life is assumed by a newly acquired product. In this definition, disposition of a possession is not a necessary condition for its replacement, and the incumbent product may be transferred to a different role in the consumer's household. For example, the purchase of a new sofa for the living room would be considered a replacement purchase even if the old sofa

is not disposed of but is instead moved into the family room. Likewise, it is not necessary for the replacement product to be an improved model or exact replica of the possession it replaces. Any newly acquired product capable of fulfilling the primary role previously occupied by an incumbent product is considered a replacement. For example, a recently-purchased SUV would be a replacement for a minivan if it replaces that vehicle's role as the main transportation vehicle for the family.

As this definition shows, the consumer replacement decision task entails not one, but two, suborder decisions. For one, the consumer must make a decision regarding a new acquisition and its capability of assuming (or surpassing) the incumbent's role in the consumer's life. The consumer must also make a disposition decision regarding an incumbent possession he or she already owns. In many cases, these suborder decisions reflect interconnected and reciprocal choices that do not necessarily represent mutually exclusive behavioral alternatives in the replacement decision task.

The replacement decision framework

When contemplating a replacement decision, three alternatives are typically available. The consumer may (1) acquire a replacement product and dispose of the incumbent possession, (2) acquire a replacement and keep the incumbent, possibly moving it to a new role in the household, or (3) engage in decision avoidance, in which the consumer postpones the decision or seeks a solution that involves no action or no change (Anderson, 2003; Dhar, 1997). Alternatives two and three allow for continued retention of the incumbent possession; alternatives one and two involve acquisition.

Several factors influence the decision outcome. The theory of planned behavior (Ajzen, 1991) holds that attitude toward the behavior, subjective norms, and perceived behavioral control determine intentions, and these intentions are translated into behavior to the extent the behavior is under the consumer's control. Accordingly, a consumer's intention to acquire a replacement product would be a function of the consumer's attitude toward acquiring a replacement, subjective norms, and ability to acquire a replacement.

However, the replacement decision process involves two subsidiary decisions—a decision about whether to acquire a replacement product and a decision about what to do with the incumbent. The central premise of our framework for consumer replacement decisions is that consideration of both decision processes leads to better prediction of consumer replacement intentions than does an assessment of the acquisition decision alone, as shown in Fig. 1. There is both theoretical and empirical support for adopting this perspective. When making a decision, an individual has a choice among alternative acts, including the option of not doing anything at all (Dhar, 1997; Wyer, 2005). The full spectrum of choice is not typically represented in attitude models, although some research suggests that incorporating attitudes toward alternative actions can improve prediction of behavioral intentions. For instance, Ajzen and Fishbein (1969) found that in a single-decision choice situation with multiple mutually exclusive alternatives, a

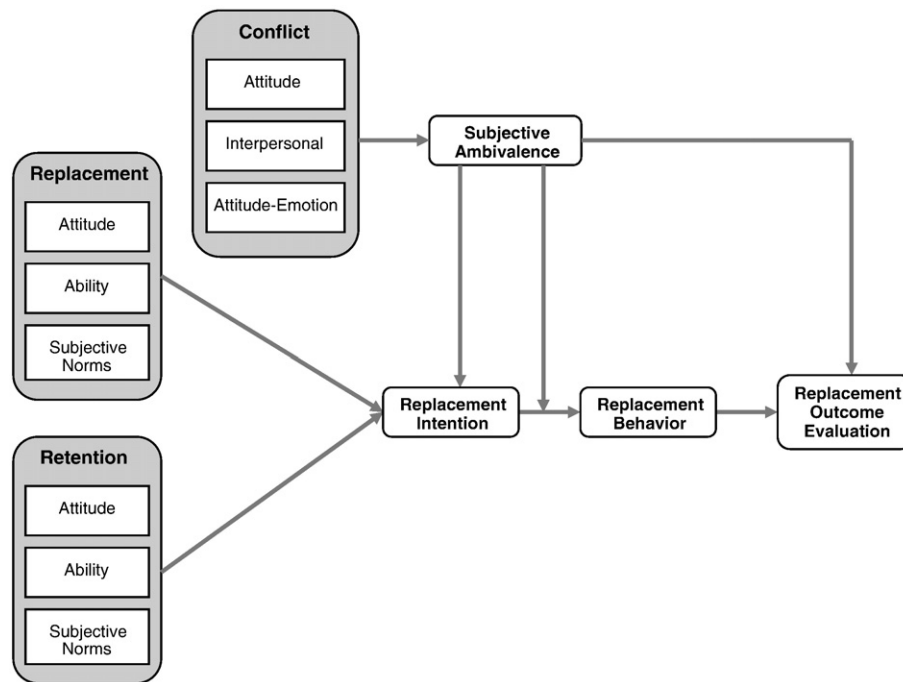


Fig. 1. Ambivalence and the expanded replacement decision model.

model that included attitudes and subjective norms toward each of the alternatives predicted intentions better than simpler models. Jaccard (1981) found that a model that included both attitude toward action and attitude toward not taking action was superior to a similar Fishbein model that included only attitude toward action in predicting behavioral intentions.

However, prior research has not investigated whether behavioral intentions can be predicted with higher accuracy by a model that includes attitudes, subjective norms, and perceived behavioral control (ability) for multiple interrelated decisions such as those encountered when considering product replacement. As stated previously, replacement decisions involve consideration of two distinct but interrelated and non-mutually exclusive decisions, one concerning acquisition and one concerning retention. We predict that a model that includes variables pertaining to both subsidiary decisions in the replacement process will be superior to a model based solely on acquisition variables, as stated in the following hypothesis:

H₁. A model that includes attitude, subjective norms, and ability variables concerning both the acquisition decision and the retention decision will predict intention to purchase a replacement product better than a model that includes only those variables related to the acquisition decision.

Ambivalence in replacement decisions

The interconnected nature of replacement subdecisions can make the decision process complex for consumers and may engender conflict, particularly when the incumbent product is in good working order. The recognition that a new product possesses desirable benefits can conflict with assessments that the incumbent product is adequate for the consumer's needs.

There may also be conflict between consumers' cognitions and the emotions they experience when contemplating replacement, as when the pleasure and joy of contemplating the purchase of a new (replacement) vehicle conflicts with one's cognitions that such a purchase might be an unwise use of the family's limited financial resources. Conflict may also arise when one's own attitudes about replacement conflict with the assessments of important others who have a stake in the decision outcome.

Thompson, Zanna, and Griffin (1995) suggest that conflicts such as these help explain why individuals sometimes feel "torn" and unsure of their attitudes when making decisions. This experience of feeling conflicted or torn is referred to as subjective ambivalence, which Priester and Petty define as "a psychological state of conflict associated with an attitude object" (Priester and Petty, 1996, p. 432).¹ In the ambivalence literature, subjective ambivalence is typically assessed by direct, self-report measures (Priester and Petty, 1996, 2001; Thompson et al., 1995). Researchers have also developed a variety of mathematically-derived indices (sometimes referred to as "objective" measures of ambivalence) that indirectly assess how feelings of psychological conflict might arise from competing reactions toward attitude objects (Priester and Petty, 2001; Thompson et al., 1995).

The literature has described at least three potential types of conflict that may be associated with ambivalence, and two of these have been investigated empirically. The first, referred to as attitude conflict or attitude ambivalence, has received the most

¹ Note that ambivalence can be distinguished from cognitive dissonance. Ambivalence involves conflicts between mental states (attitudes or affect), while dissonance involves conflict between a mental state and a behavior that has been performed by an individual. The distinction between these constructs is described in more detail by Jonas, Broemer, and Diehl (2000).

attention in the literature. Thompson et al. (1995) describe ambivalent attitudes as instances in which attitude objects are evaluated as being both positive and negative. According to Thompson et al. (1995), attitude conflict is most pronounced when competing positive and negative evaluations toward an attitude object are both intense and similar in magnitude. In extending this conceptualization to replacement decisions, it is evident that attitudes about replacement and retention may be in conflict with one another, leading to ambivalence. Thus, and consistent with prior literature, we expect to observe a positive relationship between attitude conflict and subjective ambivalence in decisions concerning product replacement.

H_{2a}. Subjective ambivalence in a product replacement decision is positively associated with conflict between one's attitudes toward replacement and toward retention.

The second form of conflict associated with subjective ambivalence that has been investigated empirically arises from interpersonal inconsistencies, that is, from differences between one's personal attitudes and those held by important others. Priester and Petty (2001) found that when college students' attitudes about a variety of topics conflict with attitudes held by their parents or other liked persons, they are more likely to experience subjective ambivalence concerning the attitude object. This is most likely to occur when the attitude object is important to the individual. We hypothesize that a similar effect will occur for conflict concerning a replacement decision.

H_{2b}. When a product replacement decision is important, subjective ambivalence concerning that decision is positively associated with conflict between one's attitude toward replacement and the attitudes of important others.

A third type of conflict has been suggested in the literature but has not been empirically examined. This potential conflict arises from the fact that stimuli often evoke an affective or emotional response as well as a cognitive one. As noted by Thompson et al. (1995), thoughts and feelings about a decision are sometimes in conflict, and when this happens, a consumer is likely to experience subjective ambivalence.

We propose that conflict between attitudes and emotions concerning a replacement decision will be associated with subjective ambivalence, but only under certain conditions. Specifically, we propose that when emotion response (either positive or negative) is low, subjective ambivalence is unlikely to occur even in the presence of conflict between attitudes and feelings. When emotion response is strong, congruence between attitudes and emotional response will be associated with low levels of subjective ambivalence, and conflict between attitudes and emotions will be associated with high levels of subjective ambivalence. These predictions are consistent with Thompson et al.'s (1995) contention that ambivalence is unlikely to occur at low levels of intensity.

H_{2c}. There is an interaction between attitude and emotion in replacement decision contexts such that subjective ambivalence is likely to occur when emotion response is strong and there is a conflict between emotion response and attitude toward

replacement. When emotional response is low or when there is congruence between emotions and attitude, subjective ambivalence will be low.

Despite the recognition that subjective ambivalence may involve multiple forms of conflict, most studies have focused on only one or two of these potential conflicts. For instance, Thompson et al. (1995), Priester and Petty (1996), Nowlis, Kahn, and Dhar (2002), and Jewell (2003) limited their analysis primarily to attitude conflict. Priester and Petty (2001) and Zemborain and Johar (2007) have investigated conflict due to interpersonal inconsistencies. And although a number of scholars have examined the notion of mixed affective responses (e.g., Lau-Gesk, 2005; Otnes et al., 1997; Williams and Aaker, 2002) as a source of conflict in consumer evaluative judgments, we are aware of no studies that have empirically examined conflicts between attitudes and emotions as a source of subjective ambivalence.

In the present study, we take a more comprehensive approach to studying the conflicts that give rise to subjective ambivalence by assessing all three potential sources of conflict and examining the relationship of these conflicts to subjective ambivalence. An additional contribution is that we tie ambivalence to behavior intentions and actual consumer purchase behaviors, something that has not been done before.

Ambivalence, intentions, and behaviors

Ambivalence is important in replacement decisions because it has the potential to influence how the decision process is carried out. We propose three possible effects, as shown in Fig. 1. First, subjective ambivalence is likely to influence replacement intentions. Particularly for durable goods, a suboptimal decision is not easily undone, and consumers may fear post-decision regret (Inman, 2007; Zeelenberg and Pieters, 2007). Consumers with ambivalence about the best course of action may play it safe by maintaining the status quo (Anderson, 2003; Dhar, 1997) or delaying the decision (Luce, 1998), as expressed in Hypothesis 3.

H₃. Subjective ambivalence about a replacement decision is negatively associated with intention to purchase a replacement and positively associated with both intention to retain the incumbent possession and an intention to delay making the replacement decision.

Second, those ambivalent consumers who do indicate an intention to replace an incumbent possession will have more difficulty in following through with that intention than will those low in ambivalence. Bagozzi and Yi (1989) observed that poorly-formed intentions are less effective predictors of behavior than well-formed intentions. Ambivalent consumers may vacillate during the decision process (Bargh, Chaiken, Govender, and Pratto, 1992; Sparks, Conner, James, Shepherd, and Povey, 2001), creating instability in their intentions. As a result, the association between a consumer's intention at one point in time and that consumer's ultimate behavior may be weaker for ambivalent consumers than for those who are not ambivalent, leading to Hypothesis 4.

H₄. The relationship between consumers' replacement intention and actual replacement behavior will be stronger for consumers with low ambivalence than for those with high ambivalence.

Third, the vacillations associated with ambivalence may influence consumers' evaluation of their ultimate decision. Delays and vacillations provide consumers with more opportunities to explore possible outcomes, both good and bad, associated with alternative courses of action and more chances to second guess. Hence, once a decision is finally made, ambivalent consumers may have lower satisfaction, lower confidence in their decision, and more decision-related negative affect. Some support for this idea is provided by [van der Pligt, Zeelenberg, van Dijk, de Vries, and Richard \(1998\)](#), who report that consumers with mixed feelings anticipate having more post-decision negative affect than do those without mixed feelings. We test whether these proposed negative affects actually occur.

H₅. Once they do make a replacement decision, consumers with high levels of ambivalence (compared to those with low levels) are likely to have

lower satisfaction with the decision,
lower confidence that they made the correct decision, and
more negative affect concerning the decision.

Hypotheses were tested using data from a mail survey of consumers currently engaged in a replacement decision for a durable possession. One year later, a follow-up survey recorded these consumers' actual behaviors and satisfaction with decision outcomes.

Methods

Sampling and survey procedures

The sample was a randomly drawn list of 1305 adult residents of a midwestern state purchased from a vendor. A two-wave mailing procedure was used. The first wave included the cover letter, a copy of the questionnaire, and a postage-paid return envelope. As an incentive, a \$1 bill was enclosed in each of the initial mailing packets, and all respondents were entered into a drawing held later for eight \$25 gift certificates from Wal-Mart. Two weeks after the initial mailing, those who had not yet responded were sent a reminder letter with another copy of the questionnaire and a return envelope. Responses were tracked by an identification number that could be removed if the respondent chose to remain anonymous.

To be included in the study, respondents must have been actively contemplating the replacement of a durable possession that was in reasonably good working condition. Instructions first asked respondents to think of things they currently owned "that last a long time" that were still in "good working order," and then asked if they were thinking of replacing any of those objects within the next year. Examples of durable goods were provided in the instructions ("cars, boats, major household appliances, TVs, stereos, etc."). Allowing for a variety of

product types ensured adequate variance in the measures and reduced product-related bias. Respondents who reported they were not currently considering replacing a durable good in the coming year and had no plans to were deemed ineligible. Qualified respondents were asked to name and describe the possession they were considering replacing and to respond to all questions in reference to their decision to replace that particular possession.

Sample characteristics and response rate

The mailing resulted in 404 responses for a response rate of 37% (404/1105 deliverable contacts). Of these, 179 cases were deemed ineligible for reasons described above. Two incomplete surveys and three cases representing multivariate outliers were removed, leaving a sample of 220 cases. Sample demographics are generally comparable with population demographics, although the sample is slightly older and more upscale than the general population. In the final sample, 54% of respondents were male, 64% were married, 52% were above the age of 45, and 32% had annual household incomes of \$75,000 or more. The most frequently mentioned candidate product for replacement was a vehicle, mentioned by 39% of respondents. Other product categories reported on were audio or video equipment, personal computers, major appliances, and furniture.

Follow-up survey

A follow-up survey was mailed one year later to 210 of the original respondents for whom there were identification numbers. The mailing included a cover letter that reminded participants of the prior survey and asked them to report further details about their replacement decision. The possession reported by the respondent in the original survey was listed at the top of each questionnaire to serve as a reminder. A \$1 token incentive was included in each mailing. Response rate was 65% (131/201 deliverable contacts). Data from three respondents were discarded because the product they originally reported on had been destroyed or stolen.

Measures—survey 1

The first survey included standard measures of attitude, ability, subjective norms, and intentions (see, e.g., [Ajzen, 1991](#); [Ajzen and Madden, 1986](#); and [Bagozzi and Warshaw, 1990](#)), as well as measures of ambivalence and intention to delay (consistent with [Anderson, 2003](#)). The intentions and ambivalence measures are described below; the remaining measures for the first survey are described in [Appendix A](#).

Intentions

Pretests identified several different behavioral intentions that a consumer might form following a replacement contemplation. These are (1) acquire a replacement product and dispose of the incumbent, (2) acquire a replacement and keep the incumbent as a backup or for a different use, (3) maintain the status quo with

respect to the possession, and (4) dispose of the incumbent possession and not acquire a replacement product.² Each of these intentions was assessed as the respondent's subjective probability of engaging in each of these behaviors in the upcoming year on a scale of zero ("no chance I will do this") to 10 ("I'm certain I will do this").

For purposes of analysis, two variables were created from this measure. The variable "intention to replace the incumbent possession" is reflected in alternatives 1 and 2 listed above. In both of these options, the incumbent is replaced with a new product; the two options differ only with respect to intentions concerning the disposition of the incumbent. Hence, for analysis purposes a respondent's value on the intention to replace variable was the higher of their values for alternatives 1 and 2.

The variable "intention to retain the incumbent possession" is reflected in alternatives 2 and 3. In both of these options, the incumbent possession is retained, although they differ in the role the incumbent possession will have in the future. For analysis purposes a respondent's value on the intention to retain variable was the higher of the values for alternatives 2 and 3.

Subjective ambivalence

Subjective ambivalence is typically assessed through a direct, self-report measurement of consumers' perceptions of their psychological experience of conflict. Measures of subjective ambivalence typically include a set of three or more statements that reflect conflicted reactions, mixed feelings, and indecisiveness toward attitude objects; respondents indicate the extent to which they experienced each on a numerical scale (Priester and Petty, 1996, 2001; Priester, Petty, and Park 2007). Scores are then summated to form a measure of subjective ambivalence. Consistent with this practice, subjective ambivalence in this study was measured with four items that assess these states. Specific items are shown in Appendix A; the summated scale had an alpha of .86.

Conflict

Three forms of conflict were assessed in the survey. Attitude conflict (often referred to as attitude ambivalence in the literature) is usually measured by creating index scores that reflect the positive and negative cognitive evaluations associated with the attitude object. The attitude conflict measure that has gained the most acceptance is the Griffin measure. Thompson et al. (1995) conclude in their review of alternative indices that the Griffin measure conforms most closely to their conceptualization of the two necessary and sufficient conditions for attitude ambivalence, which are: 1) that the two (positive and negative) attitude components are similar in magnitude; and 2) that both attitude components are of at least moderate intensity. Intensity is assessed by averaging the two components and (dis)

similarity is assessed by subtracting the absolute difference of the positive and negative components. The basic Griffin formula is as follows:

$$(P + N)/2 - |P - N| \quad (1)$$

where P = strength of positive evaluation and N = strength of negative evaluation.

The basic Griffin formula can be modified to conform to our conceptualization of replacement decisions. This involves creating an attitude conflict score based on attitude evaluations regarding the two subdecisions:

$$(A_1 + A_2)/2 - |A_1 - A_2| \quad (2)$$

where A_1 = attitude toward replacement and A_2 = attitude toward retention. Consistent with Griffin's formulation, the first term in this equation indicates the intensity of the potentially conflicting evaluations. A high value for the intensity term would occur when a particular consumer has strongly favorable attitudes toward both replacing the incumbent possession and retaining it. The second term of the equation indicates the level of (dis) similarity in magnitude of the consumer's attitudes toward replacement and retention. If a respondent highly favored both replacement and retention, dissimilarity would be low. The attitude toward replacement and attitude toward retention measures shown in Appendix A were used in the calculation of the Griffin score.

Interpersonal conflict was assessed with the procedure developed by Priester and Petty, which calculates the absolute difference between the respondent's attitude toward an attitude object and the attitude of significant others toward the same object (see also Zemborain and Johar 2007). In the present context, this measure would be represented by the absolute difference between the respondent's attitude toward replacement and the attitude of significant others concerning replacement (the subjective norm; see Appendix A).

To assess conflict between attitudes and emotional reactions concerning replacement, it was necessary to measure respondents' feelings about the possibility of replacing the incumbent possessions. Respondents were asked to rate the extent to which they felt six different emotions when they thought about replacing the possession, using a 5-point scale (1 = "very slightly or not at all" to 5 = "extremely"). Items included three positive (happy, enthusiastic, and excited) and three negative (guilty, nervous, and distressed) emotions drawn from the Consumption Emotions Scale (Richins, 1997). The assessment of attitude–emotion conflict is discussed in more detail in the results section. Finally, decision importance was also measured in the survey by two seven-point scale items asking respondents to indicate how important the replacement decision was to them and how much thought they had given to the replacement decision.

Measures—survey 2

The follow-up survey was shorter and took place one year after the first data collection. It contained measures of the

² The fourth alternative is more appropriately described as a disposition intention, rather than a replacement alternative, and was included to screen out respondents who were not contemplating a replacement decision. Respondents who assigned a probability of five or higher to this option (on the zero to 10 scale) were considered ineligible. This alternative is not discussed further in this manuscript.

Table 1
Correlation matrix of model variables, survey 1

Variable	1	2	3	4	5	6	7	8	9	10	11
1. Attitude toward replacement	–	.20	.15	.37	–.34	–.08	–.09	–.19	.28	–.19	–.31
2. Ability to replace		–	–.02	.12	–.08	.06	–.08	–.17	–.14	–.35	–.17
3. Subjective norm (replace)			–	.14	–.16	–.02	.10	.02	.15	–.04	.05
4. Intention to replace				–	–.38	–.05	–.09	–.22	.12	–.13	–.17
5. Attitude toward retention					–	.33	.39	.39	–.10	.16	.25
6. Ability to retain						–	.12	.50	–.01	–.03	.01
7. Subjective norm (retain)							–	.22	.14	.16	.26
8. Intention to retain								–	.11	.16	.26
9. Positive emotion									–	.15	.05
10. Negative emotion										–	.52
11. Subjective ambivalence											–

Note. *r* values .13 and above are significant at $p < .05$; *r* values .18 and above are significant at $p < .01$ (two tailed).

decision outcome (behavior), affect associated with the decision outcome, and decision satisfaction and confidence, as described below.

Behavior

Decision outcome was measured by a checklist that asked respondents to choose which statement best represented the replacement action they ultimately undertook. Possible outcomes were: (1) replaced the product and disposed of the incumbent, (2) replaced the product and kept the incumbent, (3) kept the product without replacement, or (4) had not made a decision yet. Of the 128 eligible respondents, 60.9% had replaced the item they reported on in study 1, 20.3% had decided to maintain the status quo with respect to the product (i.e., not replace it), 15.6% still had not made up their minds about whether the item should be replaced, and 3.1% discarded the product without replacing it.

Affect associated with decision outcomes

Respondents reported their feelings about the outcomes of their replacement decision by responding to the same six emotion descriptors (three positive and three negative) used in the first survey.

Decision satisfaction and decision confidence

Two seven-point scale items assessed respondents' overall satisfaction with decision outcomes and decision confidence.

Results

Predicting replacement intentions

Hypothesis 1 proposes that replacement intentions can be better understood by including variables relevant to both the replacement and the retention decision. Prior to testing this hypothesis, we assessed the discriminant validity of the variables concerning replacement versus those concerning retention. The correlation between attitude toward replacement and attitude toward retention was $-.34$ ($p < .01$), between ability to replace and ability to retain was $.06$ (n.s.), and between subjective norm to replace and subjective norm to retain was $.09$ (n.s.). These low correlations provided evidence of discriminant validity.

To test Hypothesis 1, the usual model advocated by Ajzen (1991) was first used to predict replacement intention. Independent variables were attitude toward replacement, ability to replace, and subjective norm concerning replacement. The correlation matrix for all model variables is shown in Table 1.

Results for the test of the traditional model are shown in the first column of Table 2. The traditional model explained 13% of the variance in replacement intentions.

If Hypothesis 1 is correct, prediction of replacement intention will be superior if variables concerning both replacement and retention are included in the model. Accordingly, replacement intentions were analyzed with an expanded regression model that included attitude, ability, and subjective norm variables for each of replacement and retention. Results are shown in the second column of Table 2.

Prediction using the expanded model was superior to that of the traditional model ($F(3, 213)$ for R^2 change = 6.68, $p < .01$). In the expanded model, both attitude toward replacement and attitude toward retention were statistically significant. The variables reflecting ability and subjective norms were not.

Results of this analysis support Hypothesis 1. When investigating the complex process of replacement decision

Table 2
Predicting replacement intentions with traditional and expanded models

	Traditional model	Expanded model
	β	β
Attitude toward replacement	.34 ^a (5.28)	.25 ^a (3.70)
Ability to replace	.06 (.86)	.05 (0.78)
Subjective norm (replace)	.09 (1.33)	.04 (0.69)
Attitude toward retention		–.33 ^a (–4.36)
Ability to retain		.07 (1.11)
Subjective norm (retain)		.05 (0.79)
Adjusted R^2	.13 ^a	.20 ^a
$F(3/213)$ for R^2 change ^b		6.68 ^a

Note. *t*-test values for β coefficients are in parentheses.

^a $p < .01$.

^b Comparison of traditional model with expanded process model, intention to replace.

Table 3
Assessment of the appropriateness of using difference scores in hypothesis testing

	Unconstrained equation						Constrained equation, Griffin score			Constrained equation, absolute difference			F_H
	X	Y	W	WX	WY	R^2	Griffin	R^2	F_C	$ X - Y $	R^2	F_C	
Attitude Conflict	-.34 **	.17 *	-.08	.08	-.03	.12 **	.24 **	.06 **	4.04 **	-.30 **	.09 **	2.33	1.03
Interpersonal Conflict	-.35 **	.11	-.09	.00	.06	.11 **	NA	NA	NA	-.21 **	.05 **	3.85 **	1.28

Note. For columns labeled X , Y , W , WX , and WY , table entries are standardized regression coefficients. For the Attitude Conflict row, X is attitude toward replacement and Y is attitude toward retention; for the Interpersonal Conflict row, X is attitude toward replacement and Y is attitude toward replacement of significant others (subjective norm). See Appendix B for explanations of W , WX , and WY . The dependent variable is subjective ambivalence. The columns labeled F_C contain F ratios for the constraints imposed by the absolute difference or Griffin score, which is equivalent to the test of difference in R^2 values for the constrained and unconstrained equations ($df=4$ and 214). The column labeled F_H contains F ratios for the test of higher order terms X^2 , Y^2 , and XY ($df=3$ and 214). Tests are based on Edwards (1994).

* $p < .05$.

** $p < .01$.

making, it is useful to consider aspects of both the retention decision and the replacement decision.

Decision ambivalence

Hypothesis 2 describes three forms of psychological conflict that may be associated with a decision-maker's subjective feelings of ambivalence. H2a proposes that attitude conflict will be positively associated with subjective ambivalence in a replacement decision situation. As noted earlier, the Griffin measure, shown in formula 1, is most commonly used to assess attitude conflict. This index calculates the absolute difference between two potentially conflicting attitude components and subtracts this difference from the intensity of the two attitude components.

Because we are applying this index in a novel way, we decided to examine statistically the appropriateness of doing so. Of greatest concern to us is the fact that the Griffin measure contains a difference score. A considerable literature addresses the conditions under which it is appropriate and inappropriate to use a difference score as an independent variable (e.g., Cronbach, 1958; Johns, 1981; Peter, Churchill, and Brown, 1993). The most thorough explications of these issues appear in Edwards (1994, 2002). Edwards describes the five major problems associated with the use of absolute difference indices³ and proposes a piecewise linear regression analysis that assesses whether difference scores may appropriately be used in a specific hypothesis test involving a specific data set. He also provides an alternative approach that can be used when the absolute difference index is inappropriate.

Attitude conflict

In our investigation of attitude conflict, we used the Edwards procedure to test the appropriateness of both the Griffin score

and a less restrictive basic absolute difference score, which is often used in studies of attitude congruence (e.g., Luo and Klohnen, 2005; Zalesny and Kirch, 1989). In part, this procedure involves comparing the variance explained by an unconstrained piecewise linear model with a more restrictive model imposing the constraints of the absolute difference model (or the Griffin measure). The piecewise linear model, its application to the Griffin score and absolute difference score models, and procedures for testing these models are described in Appendix B. Results are shown in row 1 of Table 3.

Inspection of the table reveals that the use of the Griffin score to measure attitude conflict in this data set is not warranted. Although the variance explained by the unconstrained model (Eq. A2) is significant ($R^2 = .12$, $p < .01$) and coefficients have the expected sign, neither of the coefficients for WX and WY was significant (see Appendix B). Furthermore, the variance explained by the unconstrained model was significantly greater than that explained by the constrained model (Eq. A3; $F(4/214) = 4.04$, $p < .05$), indicating that the constraints associated with the Griffin measure are not met for this analysis. The results for the less restrictive absolute difference measure are more promising in that the variance accounted for by the constrained model is not significantly less than that accounted for by the unconstrained model ($F(4/214) = 2.33$, n.s.), although the lack of significance for the WX and WY coefficient indicates that model fit is poor.

Because neither the Griffin score nor the absolute difference model is appropriate for testing H2a, this hypothesis was examined using the hierarchical regression procedure recommended by Edwards for such situations. In this analysis, the two variables used in the Griffin measure (attitude toward replacement and attitude toward retention) were entered in the first stage of a multiple regression analysis in which subjective ambivalence was the dependent variable; subsequent stages added variables representing possible nonlinear relationships and the interaction between the two attitudes.

The first stage regression yielded an $R^2 = .12$ ($p < .01$); coefficients for both attitude toward replacement ($\beta = -.25$, $p < .01$) and attitude toward retention ($\beta = .16$, $p < .05$) were significant. In subsequent stages, neither the interaction nor the nonlinear terms were significant. Hence, this analysis provides

³ These problems are that absolute difference indices (1) have lower reliability than their component scores (2) cannot be unambiguously interpreted, (3) confound the effects of their components, (4) impose constraints on the relationship between the component measures and the outcome that usually are not tested, and (5) reduce an inherently three-dimensional relationship between the component measures and the outcome to two dimensions.

no support for the hypothesis that attitude conflict is associated with subjective ambivalence in a replacement decision context. Instead, it appears that these two attitudes have an additive effect on subjective ambivalence such that the more negative one's attitudes are concerning replacement and the more positive one's attitudes are about retention, the greater the likelihood of experiencing subjective ambivalence. Possible explanations for this result are described in the discussion section.

Interpersonal conflict

Hypothesis 2b concerns the possible relationship between interpersonal conflict and subjective ambivalence. Prior research (Priester and Petty, 2001; Zembrain and Johar, 2007) has measured interpersonal attitude conflict by calculating the absolute difference between the consumer's attitude and the attitude of important others. We tested the appropriateness of using this absolute difference score in the context of the present study by again using the Edwards procedure. As before, the attitude measure was attitude toward replacement; subjective norm concerning replacement was used to represent the attitude of important others.

Inspection of Table 3 reveals that the use of the absolute difference score to test this hypothesis is not appropriate. Although the variance explained by the unconstrained model is significant ($R^2 = .11$, $p < .01$) only one of the coefficients (X , for attitude toward replacement) reached significance. Furthermore, the variance explained by the unconstrained model was significantly greater than that explained by the constrained model ($F(4/214) = 3.85$, $p < .01$), indicating that the constraints associated with the absolute difference score model are not met for this analysis (see Appendix B).

For this reason, the hierarchical regression procedure described by Edwards was also used to examine H2b. In this analysis, attitude toward replacement, subjective norm concerning replacement, and decision importance were entered in the first stage of a multiple regression analysis in which subjective ambivalence was the dependent variable. A second stage tested the significance of the interaction between subjective norm and decision importance (as predicted by H2b), and a subsequent stage tested for nonlinear and other higher order effects.

For the first stage regression, $R^2 = .13$ ($p < .01$); as before, the coefficient for attitude toward replacement was significant ($\beta = -.38$, $p < .01$). The coefficient for decision importance was also significant ($\beta = .15$, $p < .05$). However, the coefficient for subjective norm was not ($\beta = .08$, n.s.). Inclusion of the interaction between subjective norm and decision importance in the second stage showed this interaction was significant ($\beta = -.14$, $p < .05$; $F(1/215)$ for R^2 change = 4.16, $p < .05$). The interaction is shown graphically in Fig. 2. In subsequent stages, no higher order terms were significant.

This interaction can best be understood by recognizing that, on average, respondents' attitudes toward replacement were generally positive ($M = 5.35$), while the attitudes of friends and family were perceived to be significantly less so ($M = 3.79$; $t(219) = 14.9$, $p < .01$). Hence, a subjective norm opposing replacement tends to

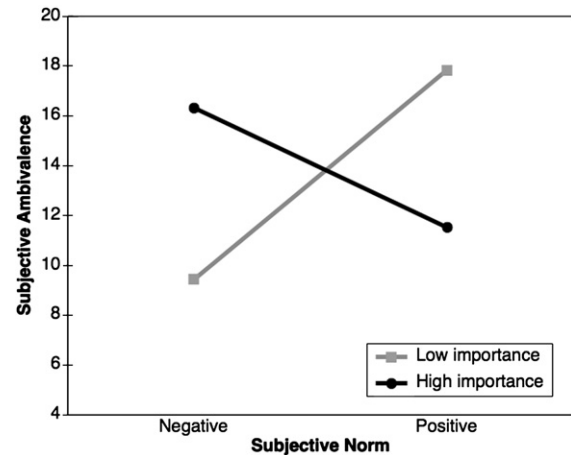


Fig. 2. Interaction between subjective norm and decision importance.

represent a conflict between the respondent and significant others. Accordingly, in Fig. 2, when the replacement decision is important to the respondent but the subjective norm is not supportive of replacement, subjective ambivalence tends to be high. On the other hand, when friends and family are seen to favor replacement in an important replacement decision, ambivalence tends to be low. Not surprisingly, ambivalence is also low when the subjective norm does not support replacement if the decision isn't very important to the consumer. Somewhat unexpected, however, is the finding that ambivalence tends to be high when the decision is low in importance but the subjective norm favors replacement. This may represent the situation in which the consumer is not motivated to replace the product, but is urged to do so by family and friends.

Attitude–emotion conflict

The third hypothesis concerning conflict, H2c, predicted an interaction between attitude and emotion in the experience of subjective ambivalence. This was tested using hierarchical regression, first in terms of positive emotions experienced when contemplating replacement and then, separately, in terms of negative emotions. Regression analysis concerning positive emotions revealed a significant interaction (F for R^2 change = 6.71, $p < .01$); the overall R^2 was .13 ($p < .01$), and all independent variables had significant predictors ($\beta_{\text{attitude}} = -.33$, $p < .01$; $\beta_{\text{positive emotion}} = .15$, $p < .05$; $\beta_{\text{interaction}} = -.17$, $p < .01$).

The interaction is graphed in Fig. 3. As predicted, ambivalence was highest when consumers reported a high level of positive emotion when contemplating replacement but simultaneously possessed unfavorable cognitions. Ambivalence was relatively low when little positive emotion was experienced or when there was congruence between consumers' emotions and attitudes toward replacement.

For negative emotions, the hierarchical regression test showed no significant interaction ($F = 1.53$, n.s.), but the coefficients for both negative emotion ($\beta_{\text{negative emotion}} = .48$, $p < .01$) and attitude were significant ($\beta_{\text{attitude}} = -.19$, $p < .01$).

Summarizing, the analyses concerning conflict and subjective ambivalence provided no support for H2a concerning attitude conflict, indirect support for H2b concerning

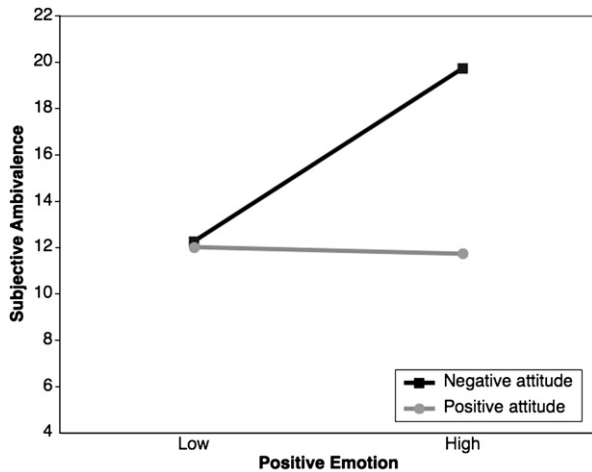


Fig. 3. Interaction between attitude and emotion.

interpersonal conflict, and support for H2c concerning the relationship between attitude–emotion conflict and ambivalence with respect to positive emotions but not for negative emotions. In addition, the tests of H2a and H2b revealed the importance of examining the extent to which the statistical assumptions underlying the use of difference scores are met before using such scores in ambivalence hypothesis tests.

Effects of ambivalence

Hypotheses 3 and 4 concern the effects of subjective ambivalence on the replacement decision process. Hypothesis 3 proposes that subjective ambivalence is negatively associated with intention to purchase a replacement and positively associated with both intention to retain the incumbent possession and intention to delay making the replacement decision. This hypothesis was supported. The correlation between subjective ambivalence and replacement intention was $-.16$ ($p < .01$). There were significant positive correlations between subjective ambivalence and both intention to retain the incumbent ($r = .26$, $p < .01$) and intention to delay ($r = .41$, $p < .01$), supporting H3.

Hypothesis 4 concerns the effects of subjective ambivalence on the replacement decision. Because ambivalence can create unstable intentions, Hypothesis 4 predicted that the association between replacement intention and behavior would be lower for respondents who exhibited pre-decision ambivalence than for those without ambivalence. This hypothesis was tested with three logistic regression analyses in which replacement behavior (did or did not replace the incumbent possession), measured on the follow-up survey, was the dependent variable. Model 1, in which intention was the only independent variable, was marginally significant ($\chi^2(1) = 3.49$, $p = .06$). In model 2, with subjective ambivalence added as an independent variable, model fit did not improve ($\chi^2(1) = .59$, n.s.). In the third model, the interaction between subjective ambivalence and intention was added to model 1. In this case, both the overall model ($\chi^2(2) = 7.40$, $p < .05$) and improvement in model fit due to the interaction variable ($\chi^2(1) = 3.91$, $p < .05$) were significant. The

relationship between intention and behavior was stronger for respondents low in ambivalence, supporting H4.

Hypotheses 5a, 5b, and 5c examined the relationship between pre-decision ambivalence and post-decision outcomes, including satisfaction, decision confidence, and negative affect. These hypotheses were tested with bivariate correlations between pre-decision ambivalence (measured in the initial survey) and the outcome variables, which were measured in the follow-up survey.

There was no relationship between subjective ambivalence and measures of decision satisfaction and decision confidence in the overall sample (satisfaction $r = -.12$, decision confidence $r = -.02$, both n.s.) or when respondents who had replaced the incumbent possession ($n = 78$) and those who had not replaced the incumbent ($n = 50$) were analyzed separately. Thus, Hypotheses 5a and H5b were not supported. However, analysis of negative affect revealed some interesting findings. For non-replacers, there was no correlation between subjective ambivalence and negative affect on any of the three descriptors (nervous, distressed, guilty). For replacers, however, there were significant relationships between ambivalence and feelings of guilt ($r = .40$, $p < .01$) and nervousness ($r = .23$, $p < .05$) about the decision. Thus, Hypothesis 5c was supported for those who decided to replace the incumbent possession, but not for those who decided against replacement.

Discussion and implications

This research highlights the complex nature of replacement decisions and proposes an expanded framework to represent the replacement process. This expanded view recognizes that replacement decisions involving durable products require consumers to resolve two possibly conflicting but related decisions: whether to acquire a replacement product and what to do with the incumbent possession. Our framework incorporates aspects of both of these interconnected but distinct decision processes into the formation of replacement intentions. We propose and demonstrate that a decision framework that takes into consideration attitudes, subjective norms, and ability concerning each of these suborder decisions can improve predictability of consumer replacement intentions.

A major contribution of this study for attitude–intention research is that it simultaneously examines attitude, ability, and subjective norms toward two non-mutually-exclusive suborder decisions to assess their potential impact on a replacement decision. Although some attitude–intention studies have demonstrated that prediction of behavior intentions can be improved by expanding traditional decision frameworks, for instance by including action and non-action as discrete alternatives (Jaccard, 1981), to our knowledge none have expanded traditional frameworks to incorporate non-mutually-exclusive choice alternatives as we have done here.

Results show that predictability of replacement intentions was significantly increased when aspects of the decision related to retention of incumbent possessions were accounted for along with those related to acquisition of a replacement product. This finding demonstrates the interconnected but independent

influence that attitude toward disposition can have on acquisition-oriented decisions and provides support for the basic premises behind our dual framework model of replacement.

This study was limited to the investigation of durable consumer goods, and future research would be useful to examine the generality of the dual framework model to a broader range of replacement decisions. Replacement decisions can encompass a wide range of products, including semi-durable goods and long-term investments such as vacation homes. The expanded framework employed in this study may be also be transferable to replacement decisions in other product categories, including decisions about changing service providers such as one's bank, physician, hairdresser, or day care provider. In addition, investigations in these broader contexts may suggest different influence variables and reveal important factors that were not included in the present study.

A second major contribution of this research is its examination of ambivalence in the replacement decision process. The dual framework model of replacement permits a closer examination of the conflict that can arise when potentially competing forces concerning two decisions are combined to create a global intention. When decisions are complex, as they often are when the consumer must simultaneously balance acquisition and disposition decisions, the potential for psychological conflict increases, creating conditions ripe for consumers to experience ambivalence. This study examined the possible influence of ambivalence on downstream variables, including intentions, purchase behavior, and evaluations of post-decision satisfaction. It also examined potential upstream variables that may be associated with ambivalence, specifically conflict. This study is the first to examine a comprehensive set of psychological conflicts that may contribute to subjective ambivalence, including attitude, interpersonal, and attitude–emotion conflict.

In terms of downstream variables, results showed that ambivalence in a replacement decision is negatively associated with an intention to replace the incumbent possession. It is positively related to both an intention to retain the incumbent and an intention to delay the decision altogether. Further, the effects of ambivalence can be of long duration. The relationship between replacement intention and actual replacement behavior, when measured one year later, was lower for ambivalent consumers than it was for those who did not experience ambivalence, indicating that ambivalence can undermine the strength of the relationship between intention and ultimate behavior.

Although we found little relationship between post-purchase consumption emotions and ambivalence measured before the purchase, we did find that ambivalent consumers who went ahead and replaced the product, despite their ambivalence, were more likely to feel guilty or nervous about their decision than were those consumers who replaced the product but were low in ambivalence. While guilt has been studied in the context of impulsive and compulsive purchasing (e.g., Faber and O'Guinn, 1992; Rook, 1987), it has rarely been studied in the context of planned purchase behavior. The relationship between ambivalence and guilt in this context deserves further investigation. For

instance, are “ambivalent replacers” more watchful for signs that they did not make the right choice? Are they more likely to return goods at the first sign of a problem or malfunction? How does making an “uneasy” choice impact future durable replacement decisions?

Another important contribution of this research is its examination of potential upstream variables associated with ambivalence, namely, conflict. We examined three forms of conflict potentially associated with ambivalence. Although all three of these conflict forms have been identified in the ambivalence literature, only one (attitude conflict) has been investigated somewhat extensively, while another (interpersonal conflict) has received a small amount empirical attention, and the third (attitude–emotion conflict) appears to have been overlooked in empirical studies.

Our investigation of potential associations between these conflicts and ambivalence forced us to confront some methodological issues concerning conflict measures. In the literature, interpersonal conflict has been measured as the absolute difference between opinions of the self and opinions of others concerning an attitude object (e.g., Priester and Petty, 2001; Zemborain and Johar, 2007); the dominant measure of attitude conflict is the Griffin score, which also incorporates an absolute difference measure. The extensive literature on problems associated with the use of difference scores in statistical tests led us to assess whether these two measures are appropriate to assess conflict in our data, particularly because we were assessing attitude conflict in a novel context.

Relying on and extending Edwards' (1994) careful analysis of difference score measures, we determined that neither the Griffin score nor the absolute difference score measures could validly be used with our data. Hence, our analysis relied on an alternative approach using hierarchical regression to measure conflict. We examined the relationship between subjective ambivalence and three possible sources of conflict: attitude, interpersonal, and attitude–emotion conflict.

Analysis found support for the proposed association between interpersonal conflict and ambivalence. When a replacement decision is important, negative subjective norms are associated with subjective ambivalence, but this is not the case for less important decisions.

The findings concerning attitude conflict, however, are more complex. Analysis revealed no evidence of an association between attitude conflict and subjective ambivalence in replacement decisions, when attitude conflict consists of conflict between attitude toward replacement and attitude toward retention. Instead, we found an additive effect on subjective ambivalence: negative attitude toward replacement and positive attitude toward retention are both associated with greater subjective ambivalence. While at first glance this result is surprising, further consideration suggests a plausible explanation.

Attitudes in this study were measured using standard items that have been used in much of the past research on attitudes and intention (e.g., favorable–unfavorable, negative–positive), but we suspect that these conventional measures omit important attitudes and values that are specifically relevant to consumption decisions. Particularly in developed Western economies,

materialism tends to be high and the desire to acquire is strong among many consumers (e.g., Cross, 2003). Remember that for all study participants, the product they were thinking of replacing was in good working order. To even contemplate replacing such an item suggests an element of desire for something new. So it is likely that even those respondents with an unfavorable attitude toward replacement (e.g., who believed it would be unwise) still wished they could have a new one, and as a result experienced conflict. But rather than attitudinal conflict, as assessed by conventional attitude conflict measures (often referred to as “objective ambivalence” in the literature), the conflict may have been between attitudes (what would be best for the consumer) and desires (what the consumer wants). Indeed, there is some evidence for this explanation when one examines the findings concerning attitude–emotion conflict. When strong positive emotions are associated with replacement contemplation, a negative attitude toward replacement is associated with subjective ambivalence, suggesting that this form of conflict may indeed lead to subjective ambivalence.

These findings indicate that subjective ambivalence may be more complex than originally thought. Future research is needed to confirm these findings and to assess the generalizability of the complexities revealed here, particularly with respect to conflicts between attitudes and desires. Future research would do well to include nuanced measures of conflict and ambivalence that will allow these complexities to be revealed. Furthermore, the decision process itself may be more complex than conceptualized in this study. Consumer replacement decisions may sometimes evoke a more complex chain of considerations. For instance, if a consumer decides to keep an old living room sofa after it is replaced, does that sofa then replace grandma’s sofa in the den or are both old sofas retained? A society of abundance has contributed to a growing number of hoarders and packrats who continue to acquire replacement goods but experience high levels of conflict when it comes to disposing of still-usable goods.

The fourth major contribution of this research is that it describes methodological considerations that should accompany the use of measures that rely on absolute discrepancy scores when measuring sources of conflict that may contribute to subjective ambivalence. Currently accepted measures based on absolute difference scores may not always be appropriate, given characteristics of a particular data set. In our case, the data did not support use of these popular measures. We extended the procedures described by Edwards for absolute difference scores to the more specific case of the Griffin measure and describe a series of procedures that a researcher can use to ascertain if a data set meets the conditions outlined by Edwards (1994). We also describe the alternative procedures that are most appropriate to use when the analysis of difference scores is inappropriate.

In conclusion, this research emphasizes the complexity of the product replacement process. Replacement decisions are often intricately tied to disposition choices, forcing consumers to deal with non-mutually exclusive alternatives and to negotiate various sorts of conflict that can emerge from a variety of sources as they attempt to navigate the decision process. Complexities such as these present many fruitful avenues for

future research to explore more deeply and precisely the role of competing alternatives and that of subjective ambivalence in directing and predicting behavior and decision outcomes.

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Appendix A. Variable in first survey

Variable	Response scale
<i>Attitude toward replacement</i> ($\alpha = .85$) Think about BUYING a new product to use in place of the possession you have now. How would you describe your overall attitude about buying a new product?	Four 7-point semantic differential response scales anchored by negative/positive, unfavorable/favorable, foolish/wise, and harmful/beneficial
<i>Attitude toward retention</i> ($\alpha = .89$) Think about KEEPING the possession you currently own, to use as you do now or in some other way. How would you describe your overall attitude about keeping this possession?	Same as above
<i>Ability to replace</i> Does anything prevent you from being able to purchase a new replacement product within the next year?	5-point response scale anchored by “definitely yes” and “definitely no”
<i>Ability to retain</i> If you were to purchase a new product, would you be able to keep the possession you have?	Same as above
<i>Subjective norm (replace)</i> My friends want me to replace what I have now with a newer product. My family wants me to replace what I have now with a newer product.	5-point Likert scale (transformed to 7-point scale for analysis)
<i>Subjective norm (retain)</i> My friends want me to continue to keep this possession. My family wants me to continue to keep this possession.	Same as above
<i>Intention to delay decision (likelihood)</i> Continue to search and think about it some more. Put the decision off altogether for the time being.	scale numbered 0 to 10 with anchors “no chance I will do it” and “I’m certain I will do it”
<i>Ambivalence</i> ($\alpha = .86$) Sometimes I have mixed feelings about what to do when I think about replacing this possession. I can’t make up my mind one way or another about what is the best course of action for me to take.	7-point response scale anchored by “not at all true for me” and “completely true for me”; score range is 4–21

(continued on next page)

Appendix A (continued)

Variable	Response scale
<i>Ambivalence (alpha = .86)</i>	
I sometimes find myself feeling “torn” between thoughts of continuing to keep this possession or replacing it with a new product.	
When it comes to replacing this possession, my mind tells me one thing but my heart tells me another.	

Appendix B. Testing the Appropriateness of Measures Incorporating Difference Scores: An Extension to the Griffin Measure*B.1. Absolute difference measure (Edwards, 1994)*

The relationship $Z = f |X - Y|$ can be represented in the following regression equation:

$$Z = b_0 + b_1(1 - 2W)(X - Y) + e \quad (A1)$$

where $W = 0$ if $X \geq Y$ and $W = 1$ if $X < Y$. Expanding this equation, the general piecewise linear equation to test the implied constraints of an absolute difference measure of congruence for variables X and Y is as follows (Edwards 1994, p. 61):

$$Z = b_0 + b_1X + b_2Y + b_3W + b_4WX + b_5WY + e \quad (A2)$$

with the following implied constraints:

- 1) the coefficients on X and Y are equal in magnitude but opposite in sign;
- 2) the coefficients on WX and WY are equal in magnitude but opposite in sign;
- 3) the coefficient on WX is twice as large as the coefficient on X but opposite in sign; and
- 4) the coefficient on W is zero.

Further, X and Y must be measured on equivalent scales (i.e., share the same origin and interval size) at an interval or ratio level of measurement and be centered on the same value (normally the midpoint of the shared scale), and the value of b_3 must not differ from zero on all points along the line $X = Y$ in a 3 dimensional space defined by X , Y , and Z (see Edwards 1994, p. 63, for the appropriate procedure for testing this requirement).

B.2. Extension to the Griffin measure

The Griffin measure contains two terms: one representing the absolute difference between the X and Y variables and the other representing the average magnitude of the X and Y variables, in the form $(X + Y)/2 - |X - Y|$. The relationship $Z = f ((X + Y)/2 - |X - Y|)$ can be represented in the following regression equation:

$$Z = b_0 + b_1((X + Y)/2 - (1 - 2W)(X - Y)) + e \quad (A3)$$

The piecewise linear equation to test the implied constraints of this index takes the same form as the test of the absolute difference equation (formula A2), but the implied constraints differ. They consist of the following:

- 1) the coefficients on X and Y are opposite in sign;
- 2) the magnitude of the coefficient on X is not significantly different than 3 times the magnitude of the coefficient on Y ;
- 3) the coefficients on WX and WY are equal in magnitude but opposite in sign;
- 4) the magnitude of the coefficient on X is not significantly different than 4 times the magnitude of the coefficient on WX but opposite in sign;
- 5) the magnitude of the coefficient on Y is not significantly different than 75% of the magnitude of the coefficient on WY but opposite in sign; and
- 6) the coefficient on W is zero.

B.3. Testing procedure

Tests to evaluate either of these models requires that the following be established:

- 1) the proportion of variance explained by equation A2 is significant;
- 2) all coefficients in A2 except b_3 are significant and in the appropriate direction (as described in the constraints associated with the model);
- 3) the implied constraints are valid; and
- 4) no higher-order (nonlinear) terms are significant.

The validity of the implied constraints (step 3) can be assessed simultaneously by testing the difference between the multiple correlations yielded by equations A2 and A1 when testing an absolute difference model, or between equations A2 and A3 when testing a Griffin index model. If the multiple correlation obtained by equation A2 is significantly higher than the multiple correlation obtained from the alternative model, the constraints are not valid. (See Edwards 1994, p. 62, for other methods to test the implied constraints.)

If the implied constraints of the model do not hold or other conditions are not met, the index cannot be validly used in hypothesis or other tests. When this occurs, Edwards (1994, 2002) suggests that exploratory analysis involving hierarchical polynomial regression be carried out by first entering the X and Y variables into an equation, then adding sets of terms representing models of progressively higher order until no significant additional variance is explained. Because of the exploratory nature of this alternative approach, conclusions concerning results must be considered tentative and require both conceptual scrutiny and additional validating evidence.

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